

Figure report

Generated 10-Sep-2026 10:49:41 on R5611351, MATLAB 26.1.0.3312084 (R2026a) Update 4.

Figure root	C:\Users\m218089\Desktop\github_repos\FractionalReservoir\
Sections	20
Images	26
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Run directory	C:\Users\m218089\Desktop\github_repos\FractionalReservoir\
Preset	celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond
Run mode	medium
Result	20/20 succeeded, 13/13 in-paper
Elapsed	5.3 min

Values above are echoed from [manifest.md](#), which is the record of the run.

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doc_tables

Inlined from [doc_tables/adaptation_conditions.md](#).

Adaptation conditions

Preset: celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25 · **Model class:** SRNNCellTypePairs

GENERATED by `write_manuscript_tables` from a model BUILT from that preset under the `sfa3_std2` condition. Do not edit by hand – every value here is read off the object, so it cannot disagree with what the sweeps ran. Regenerate after changing the preset.

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Every sweep runs each grid point under all 3 regimes, on the **same network** (the weight seed is shared), so the comparison is paired.

Condition	SFA timescales τ_a (s)	Depressing routes	Facilitating routes
No Adaptation	none	none	none
Single-Timescale Adaptation	0.25	EE, EI, IE, II	none
Multiple-Timescale Adaptation	0.25, 1.581, 10	EE, EI, IE, II	none

Inlined from [doc_tables/adaptation_conditions_table.md](#).

Condition	SFA timescales τ_a (s)	Depressing routes	Facilitating routes
No Adaptation	none	none	none
Single-Timescale Adaptation	0.25	EE, EI, IE, II	none
Multiple-Timescale Adaptation	0.25, 1.581, 10	EE, EI, IE, II	none

Inlined from [doc_tables/equation_table.md](#).

SRNN model equations

Preset: celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25 · **Model class:** SRNNCellTypePairs

GENERATED by `write_manuscript_tables` from a model BUILT from that preset under the `sfa3_std2` condition. Do not edit by hand – every value here is read off the object, so it cannot disagree with what the sweeps ran. Regenerate after changing the preset.

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System equations

$$dx_i = \frac{-x_i + u_i + \sum_{j=1}^N w_{ij} \theta_j}{\tau_d} dt + \frac{\sigma_u}{\tau_d} dW_i$$

$$\theta_i = r_i \prod_{m=1}^M b_{im}$$

$$r_i = \phi \left(x_i - a_{0_i} - \frac{c}{K} \sum_{k=1}^K a_{ik} \right)$$

$$\frac{da_{ik}}{dt} = \frac{-a_{ik} + r_i}{\tau_{a_k}}, \quad k = 1, \dots, K$$

$$\frac{db_{im}}{dt} = \frac{1 - b_{im}}{\tau_{rec_m}} - \frac{b_{im} r_i}{\tau_{rel_m}}, \quad m = 1, \dots, M$$

Additive Wiener noise enters **only** x , which keeps the diffusion constant: Ito and Stratonovich coincide, the Milstein term vanishes, the variational equation is untouched, and the noise cancels in Benettin’s trajectory difference. σ_u is input-referred, so it is comparable to the stimulus amplitude.

ϕ is centred on zero; a_{0_i} is the **setpoint** (`S_c`), drawn per neuron as $a_{0_i} = 0.2 + \sigma_{S_c} \xi_i$, $\xi_i \sim \mathcal{N}(0, 1)$, with $\sigma_{S_c} = [0.1 \ 0.1]$ per cell type. K is the number of SFA timescales and M the number of depression timescales; their values for this preset are in the parameter table below.

The rate r_i is the **pre-depression** output of the nonlinearity. Depression and facilitation enter presynaptically, as factors multiplying r_j inside the recurrent sum, so both mechanisms are driven by the raw rate r_i and not by the depressed output.

θ_i is the **synaptic output**: the rate after depression, and the quantity the recurrent sum transmits.

The depression timescales enter as a **product**, which is the reason for using more than one: 2 of them SQUARE the depression at steady state, so the extra timescale buys DEPTH and not merely a slower approach. There is no split of τ that would make them match a single timescale.

With $\tau_{rec}/\tau_{rel} = 8$ on every timescale, each factor relaxes to $b_k(r) = 1/(1 + 8r)$, so at $r = 1$ a single timescale gives a gain of 0.1111 (a 9-fold reduction against an undepressed synapse) while 2 give 0.01235 – a **81-fold** reduction, exactly the square.

Adaptation, by contrast, enters as a **sum**, and is normalised by its timescale count: the rate subtracts $(c/K) \sum_{k=1}^K a_k$. Since every a_k relaxes to the rate, $\sum_k a_k \rightarrow Kr$ regardless of the τ_k , so the steady state is cr whatever K is – adding SFA timescales changes the route, not the destination. c is therefore the **total** adaptation budget. No such normalisation applies to depression, which multiplies rather than sums.

The sparse weights are drawn element-wise and scaled by the synaptic gain g :

$$w_{ij} = g S_{ij} (\tilde{\sigma}_{ij} \xi_{ij} + \tilde{\mu}_{ij}), \quad \xi_{ij} \sim \mathcal{N}(0, 1), \quad S_{ij} \sim \text{Bernoulli}(\alpha)$$

with $(\tilde{\mu}, \tilde{\sigma})$ indexed by (**postsynaptic**, **presynaptic**) cell type, and given as multiples of $F = 1/\sqrt{n\alpha(2-\alpha)}$.

Parameters as run

Symbol	Name	Value	Units
n	Network size	500	neurons
α	Connection probability	0.2	–
indegree	Expected in-degree	100	synapses
f	Fraction per cell type	[0.5 0.5]	–
τ_d	Dendritic time constant	0.1	s
g	Synaptic gain	1	–
$\tilde{\mu}$	Weight means (post, pre)	[8.25 -8.25; 8.25 -8.25]	multiples of F
$\tilde{\sigma}$	Weight std devs (post, pre)	[1.5 1.5; 1.5 1.5]	multiples of F
ϕ	Nonlinearity	piecewise	–
S_c	Nonlinearity setpoint	0.2	–
σ_{S_c}	Setpoint std per cell type	[0.1 0.1]	–
S_a	Piecewise slope parameter	0.8	–
K	SFA timescales per type	[3 3]	–
τ_a (E)	SFA time constants	[0.25 1.581 10]	s
τ_a (I)	SFA time constants	[0.25 1.581 10]	s
c	SFA coupling per type	[0.5 0.5]	–
M	STD timescales	2	–
τ_{rec} (E to E)	STD recovery	[2 4]	s
τ_{rel} (E to E)	STD release	[0.25 0.5]	s
τ_{rec} (E to I)	STD recovery	[2 4]	s
τ_{rel} (E to I)	STD release	[0.25 0.5]	s
τ_{rec} (I to E)	STD recovery	[2 4]	s
τ_{rel} (I to E)	STD release	[0.25 0.5]	s
τ_{rec} (I to I)	STD recovery	[2 4]	s
τ_{rel} (I to I)	STD release	[0.25 0.5]	s
σ_u	Input-referred noise	0.025	input units
σ_x	Stationary std of x	0.0559017	–
f_s	Sample rate	400	Hz

Inlined from [doc_tables/parameters_table.md](#).

Symbol	Name	Value	Units
n	Network size	500	neurons
α	Connection probability	0.2	–
indegree	Expected in-degree	100	synapses
f	Fraction per cell type	[0.5 0.5]	–
τ_d	Dendritic time constant	0.1	s
g	Synaptic gain	1	–
$\tilde{\mu}$	Weight means (post, pre)	[8.25 -8.25; 8.25 -8.25]	multiples of F

Symbol	Name	Value	Units
$\tilde{\sigma}$	Weight std devs (post, pre)	[1.5 1.5; 1.5 1.5]	multiples of F
ϕ	Nonlinearity	piecewise	—
S_c	Nonlinearity setpoint	0.2	—
σ_{S_c}	Setpoint std per cell type	[0.1 0.1]	—
S_a	Piecewise slope parameter	0.8	—
K	SFA timescales per type	[3 3]	—
τ_a (E)	SFA time constants	[0.25 1.581 10]	s
τ_a (I)	SFA time constants	[0.25 1.581 10]	s
c	SFA coupling per type	[0.5 0.5]	—
M	STD timescales	2	—
τ_{rec} (E to E)	STD recovery	[2 4]	s
τ_{rel} (E to E)	STD release	[0.25 0.5]	s
τ_{rec} (E to I)	STD recovery	[2 4]	s
τ_{rel} (E to I)	STD release	[0.25 0.5]	s
τ_{rec} (I to E)	STD recovery	[2 4]	s
τ_{rel} (I to E)	STD release	[0.25 0.5]	s
τ_{rec} (I to I)	STD recovery	[2 4]	s
τ_{rel} (I to I)	STD release	[0.25 0.5]	s
σ_u	Input-referred noise	0.025	input units
σ_x	Stationary std of x	0.0559017	—
f_s	Sample rate	400	Hz

fig_adaptation_methods

fig_adaptation_methods/Fig_single_neuron_SFA_STD.png

fig_adaptation_methods_stf

fig_adaptation_methods_stf/Fig_single_neuron_SFA_STD_STF.png

fig_dc_lle

fig_dc_lle/fig_dc_lle.png

fig_EI_param_space

fig_EI_param_space/Fig_EI_ParamSpace.png

fig_EI_weights_param_space

fig_EI_weights_param_space/Fig_EI_Weights_ParamSpace.png

fig_eig_heatmap

fig_eig_heatmap/fig_eig_heatmap.png

fig_energy_landscape

fig_energy_landscape/panelA_energy_landscape.png

fig_example_timeseries

fig_example_timeseries/fig_example_timeseries.png

fig_FI_curve

fig_FI_curve/Fig_FI_curve.png

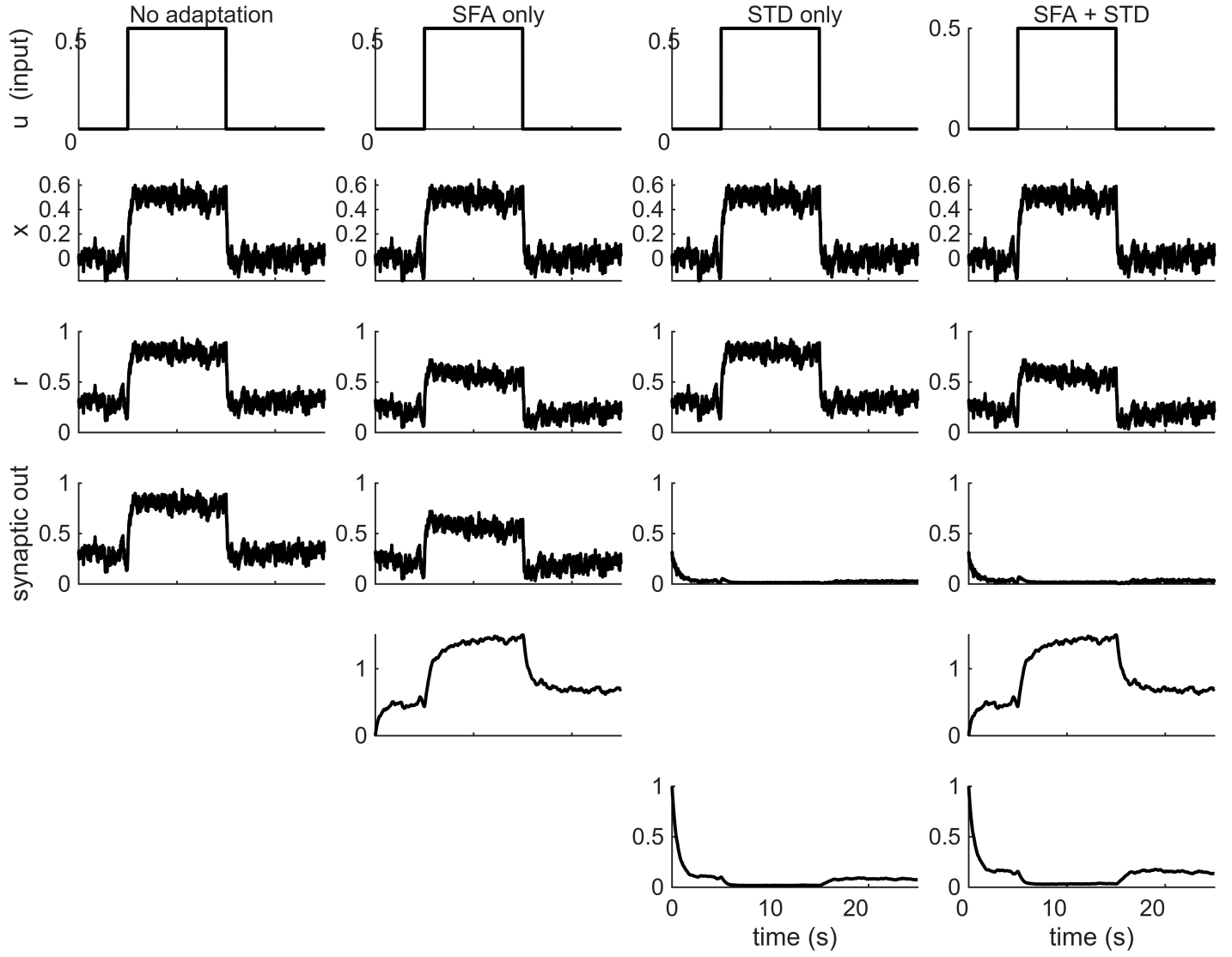


Figure 1: fig_adaptation_methods/Fig_single_neuron_SFA_STD.png

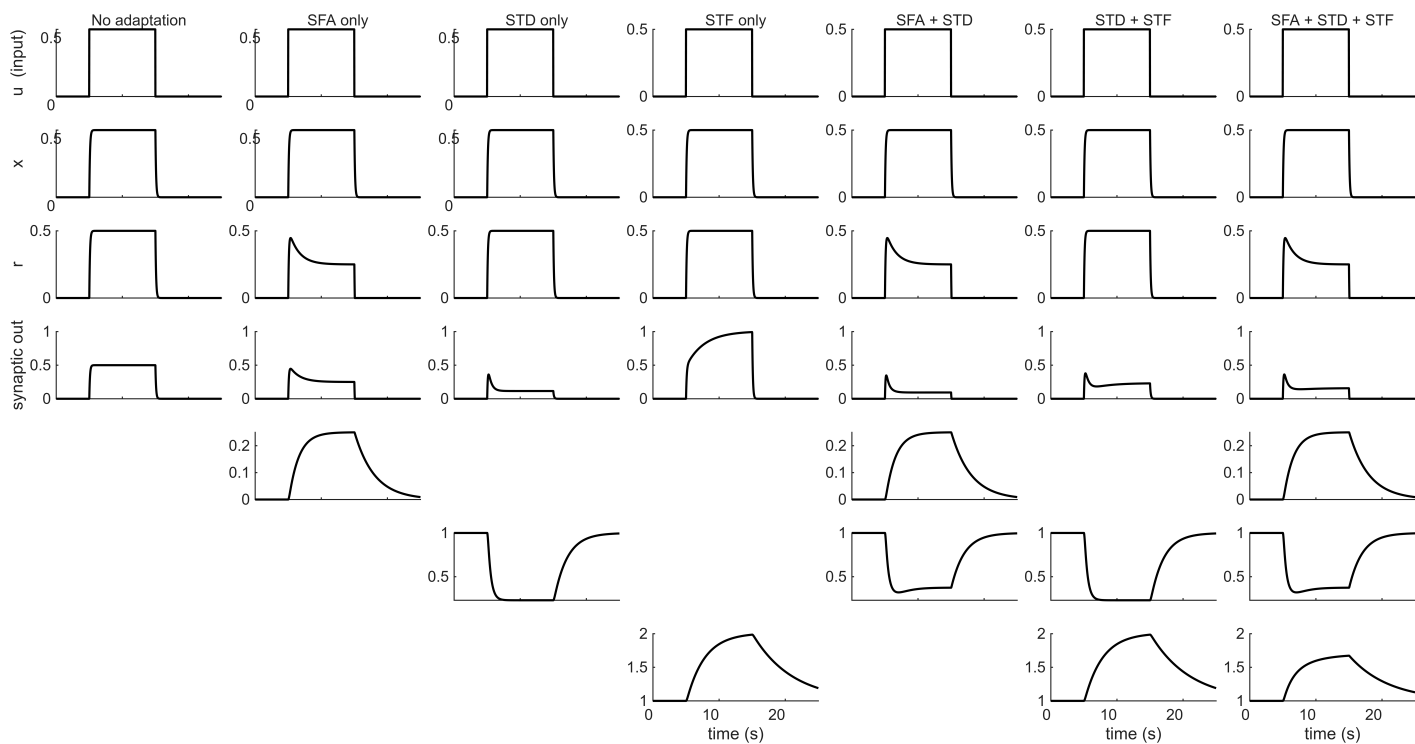


Figure 2: fig_adaptation_methods_stf/Fig_single_neuron_SFA_STD_STF.png

unov exponent vs tonic DC drive (15 seeds, celltype_pairs_sfaEl_Sc0p2sig0p1_noise0p025_dualStd_3)

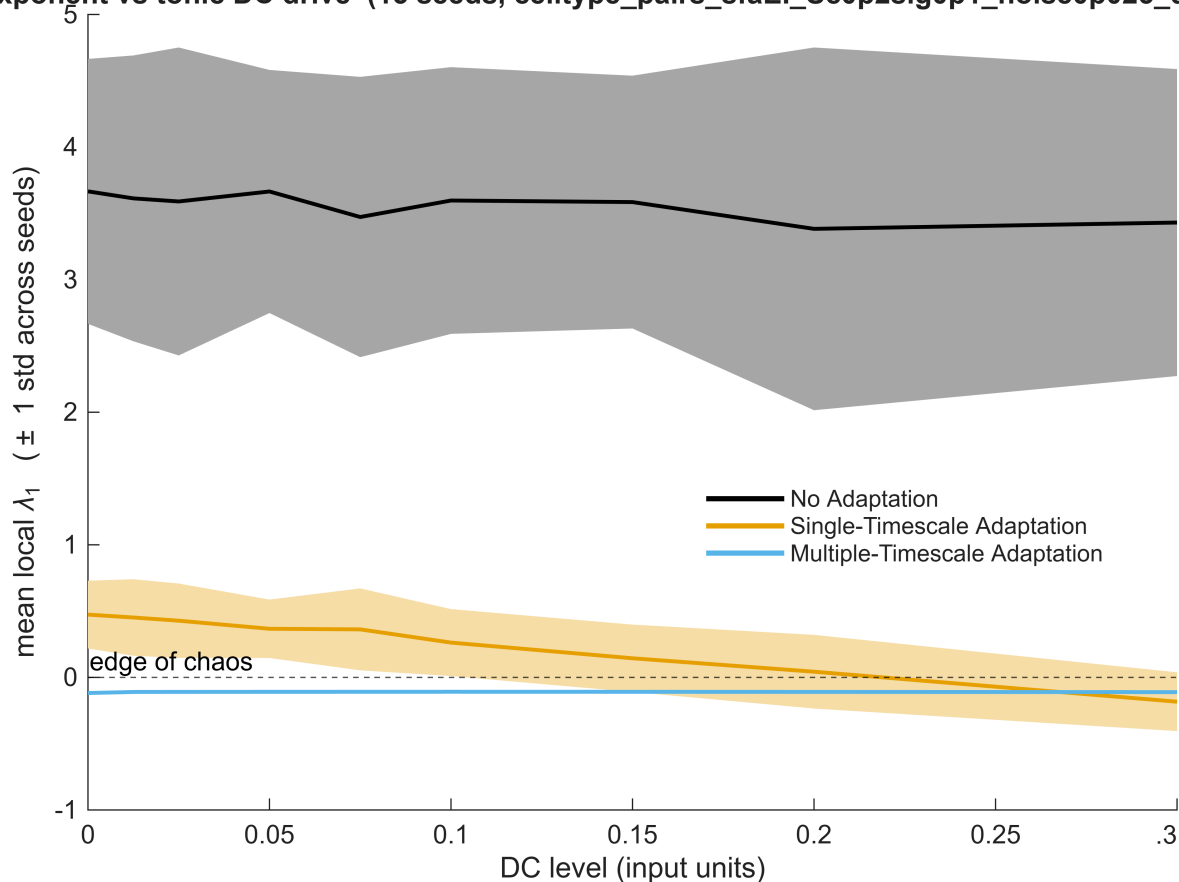


Figure 3: fig_dc_lle/fig_dc_lle.png

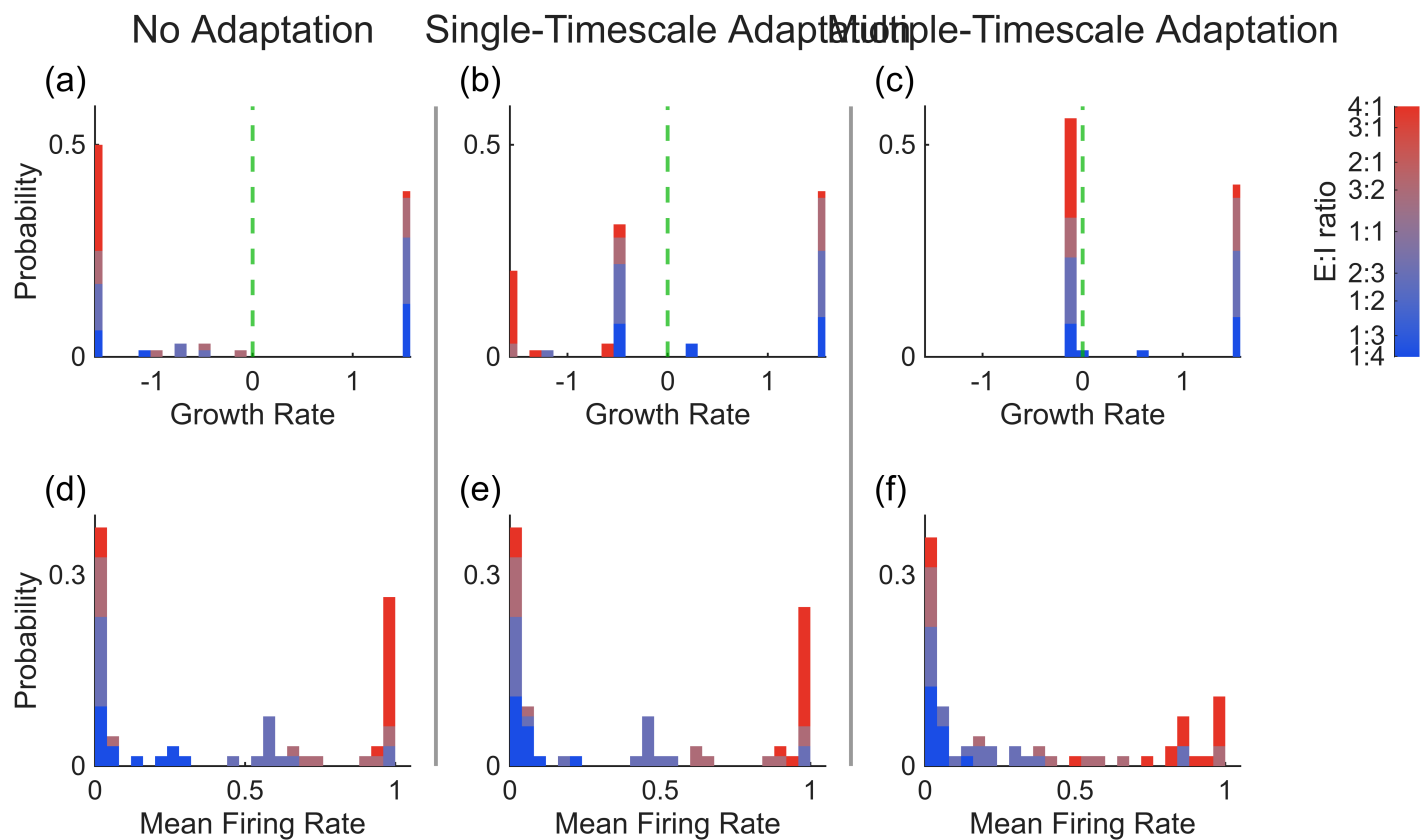


Figure 4: fig_EI_param_space/Fig_EI_ParamSpace.png

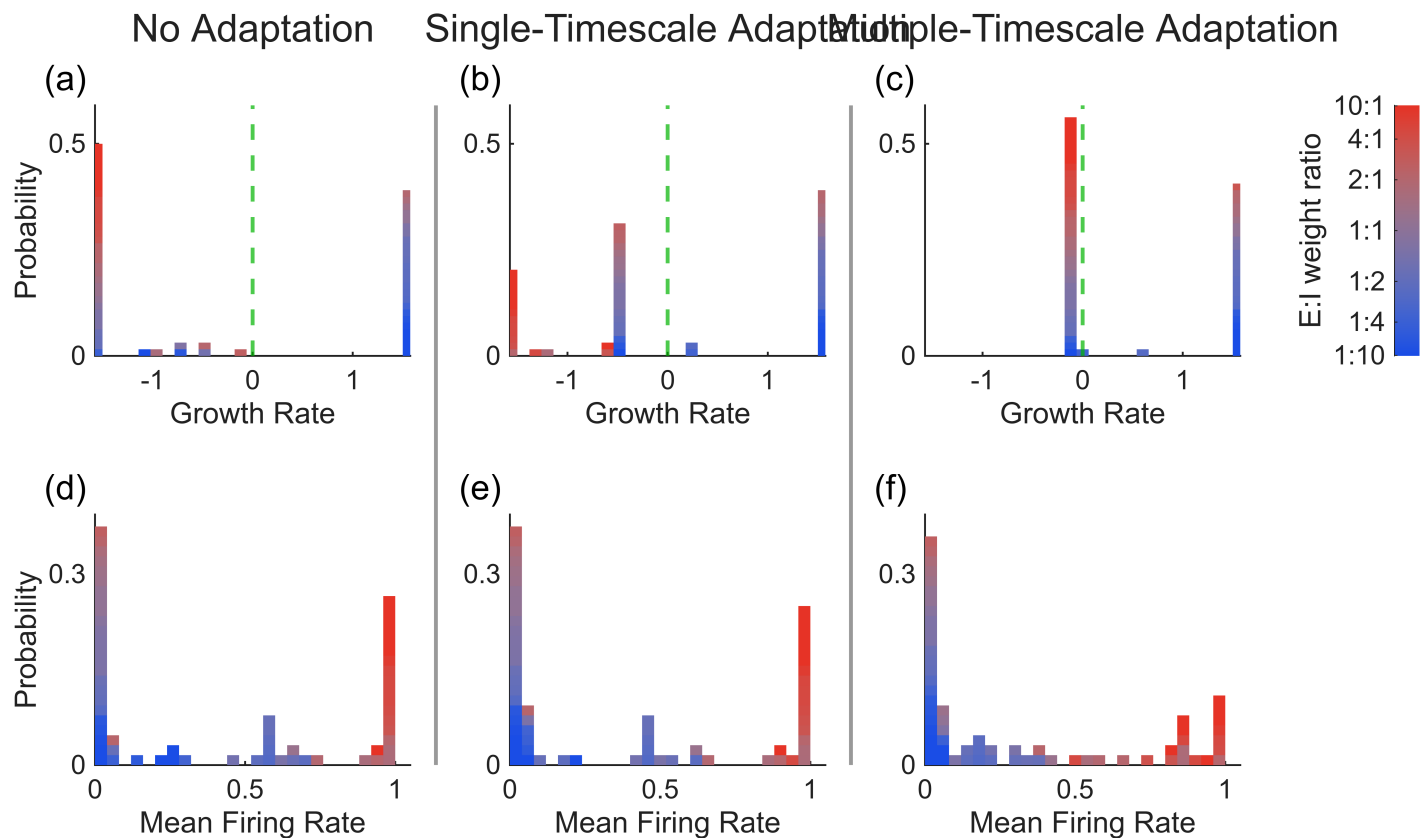


Figure 5: fig_EI_weights_param_space/Fig_EI_Weights_ParamSpace.png

Jacobian eigenvalue occupancy across adaptation regimes
LLE = finite-time Benettin exponent over the last 30 s

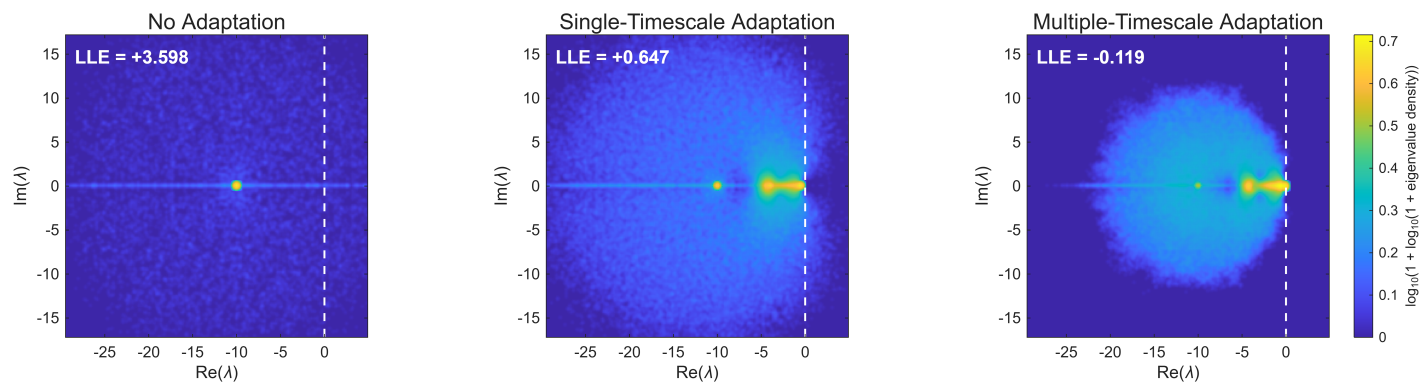


Figure 6: fig_eig_heatmap/fig_eig_heatmap.png

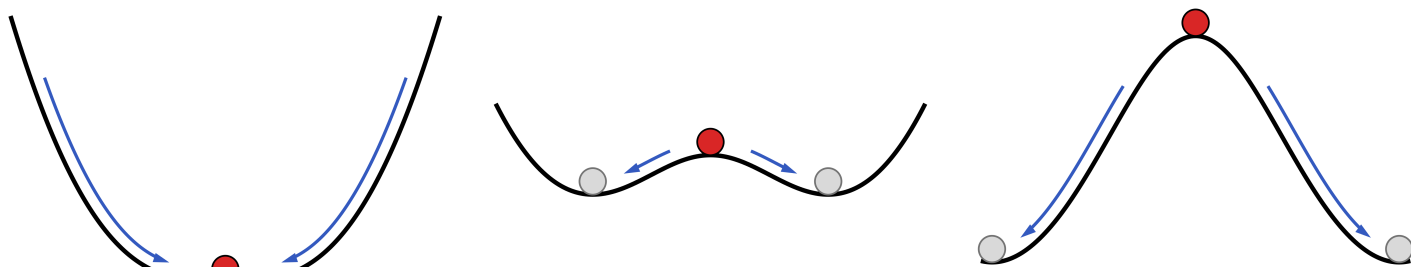


Figure 7: fig_energy_landscape/panelA_energy_landscape.png

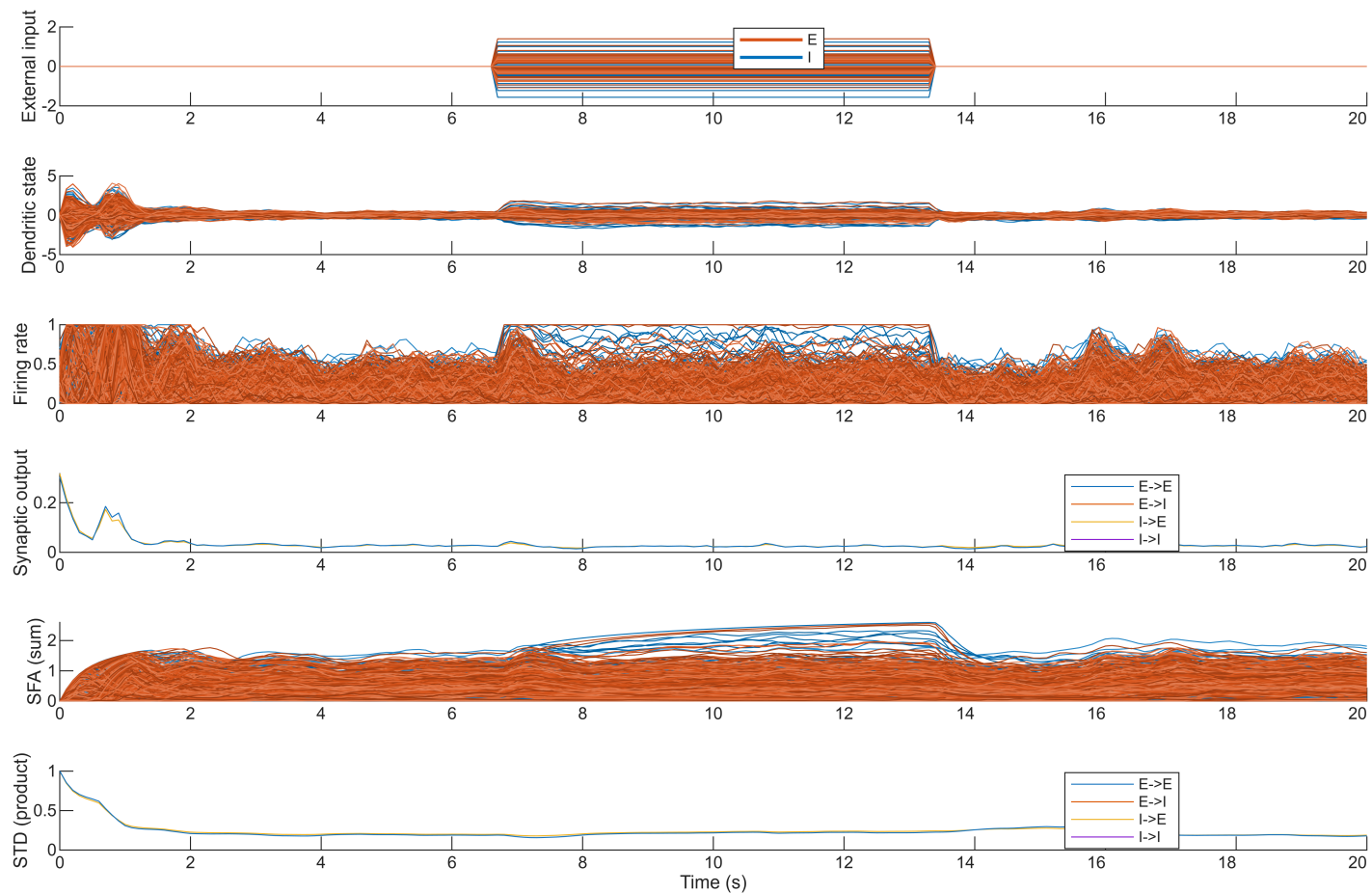


Figure 8: fig_example_timeseries/fig_example_timeseries.png

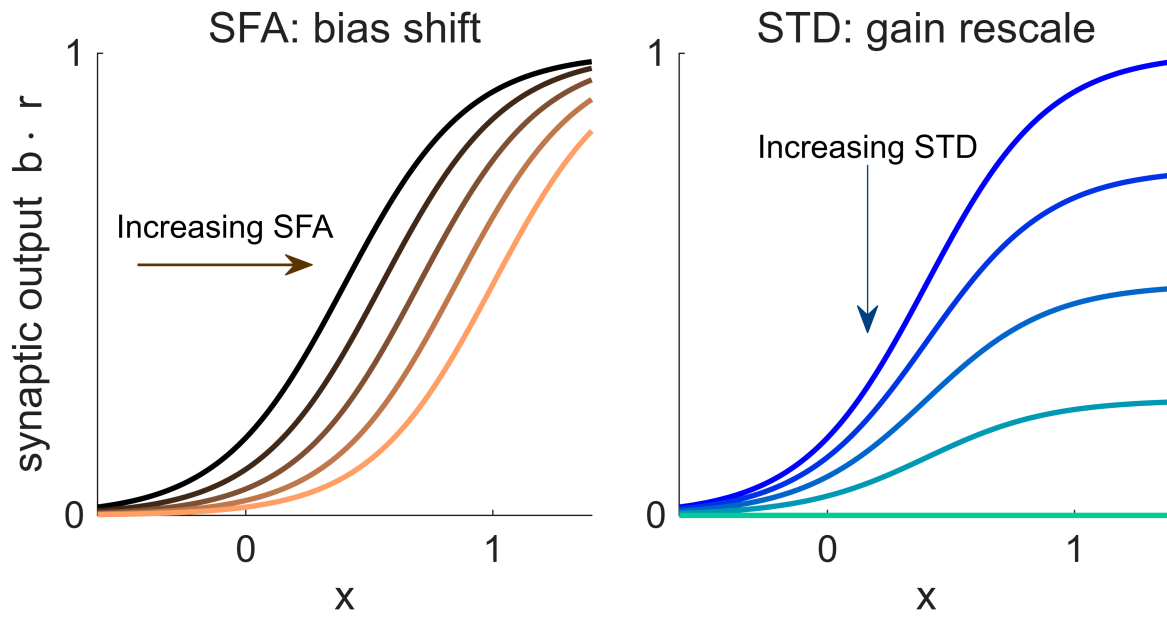


Figure 9: fig_FI_curve/Fig_FI_curve.png

fig_introduutory_concepts

fig_introduutory_concepts/eigenspectra/panelA_eigenspectrum.png

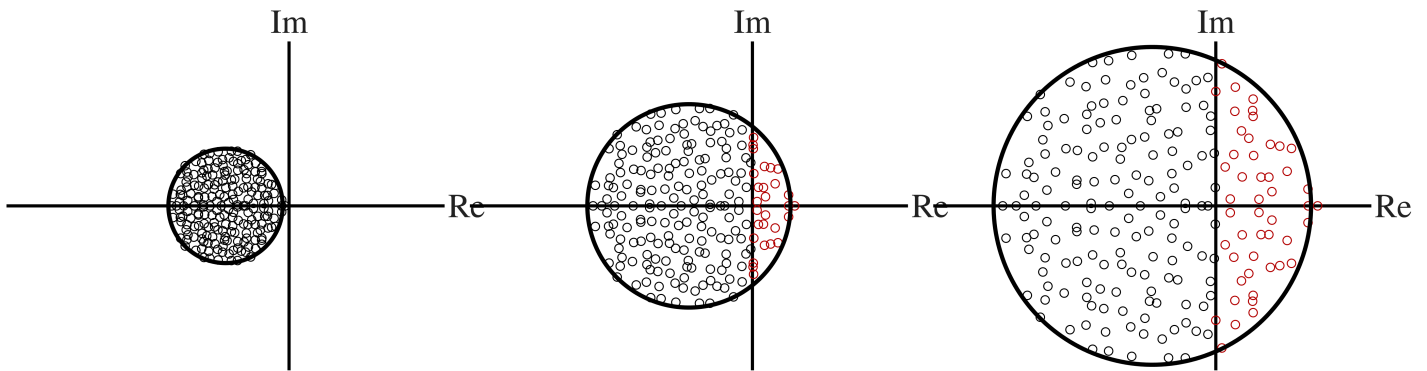


Figure 10: fig_introduutory_concepts/eigenspectra/panelA_eigenspectrum.png

fig_introduutory_concepts/statetraces/panelA_bottom_traces.png

fig_memory_capacity

fig_memory_capacity/Fig_Memory_Capacity.png

fig_memory_capacity_example

fig_memory_capacity_example/Fig_MC_Example.png

fig_param_space_allStd

fig_param_space_allStd/Fig_ParamSpace_allStd.png

fig_sensitivity_analysis_allStd

fig_sensitivity_analysis_allStd/Fig_Sensitivity_LLE_core.png

fig_sensitivity_analysis_allStd/Fig_Sensitivity_LLE_mu.png

fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_core.png

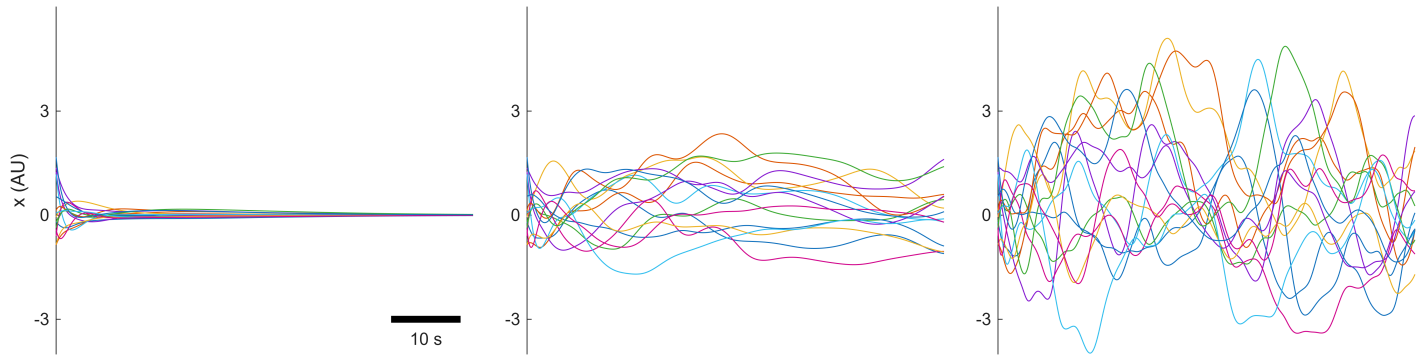


Figure 11: fig_introduutory_concepts/statetraces/panelA_bottom_traces.png

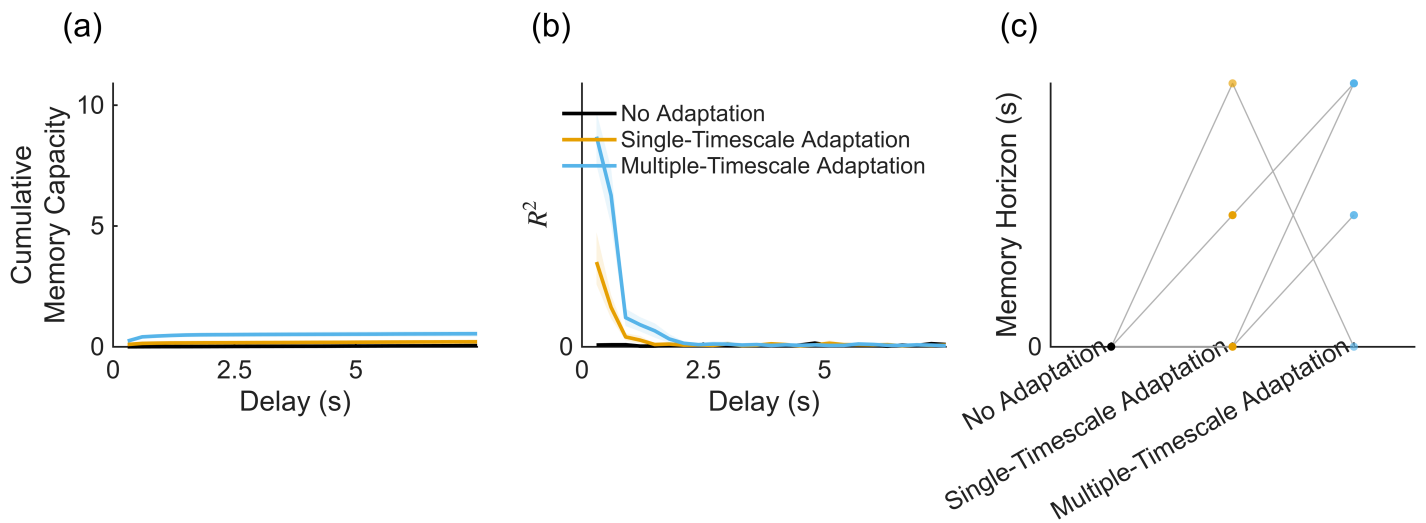


Figure 12: fig_memory_capacity/Fig_Memory_Capacity.png

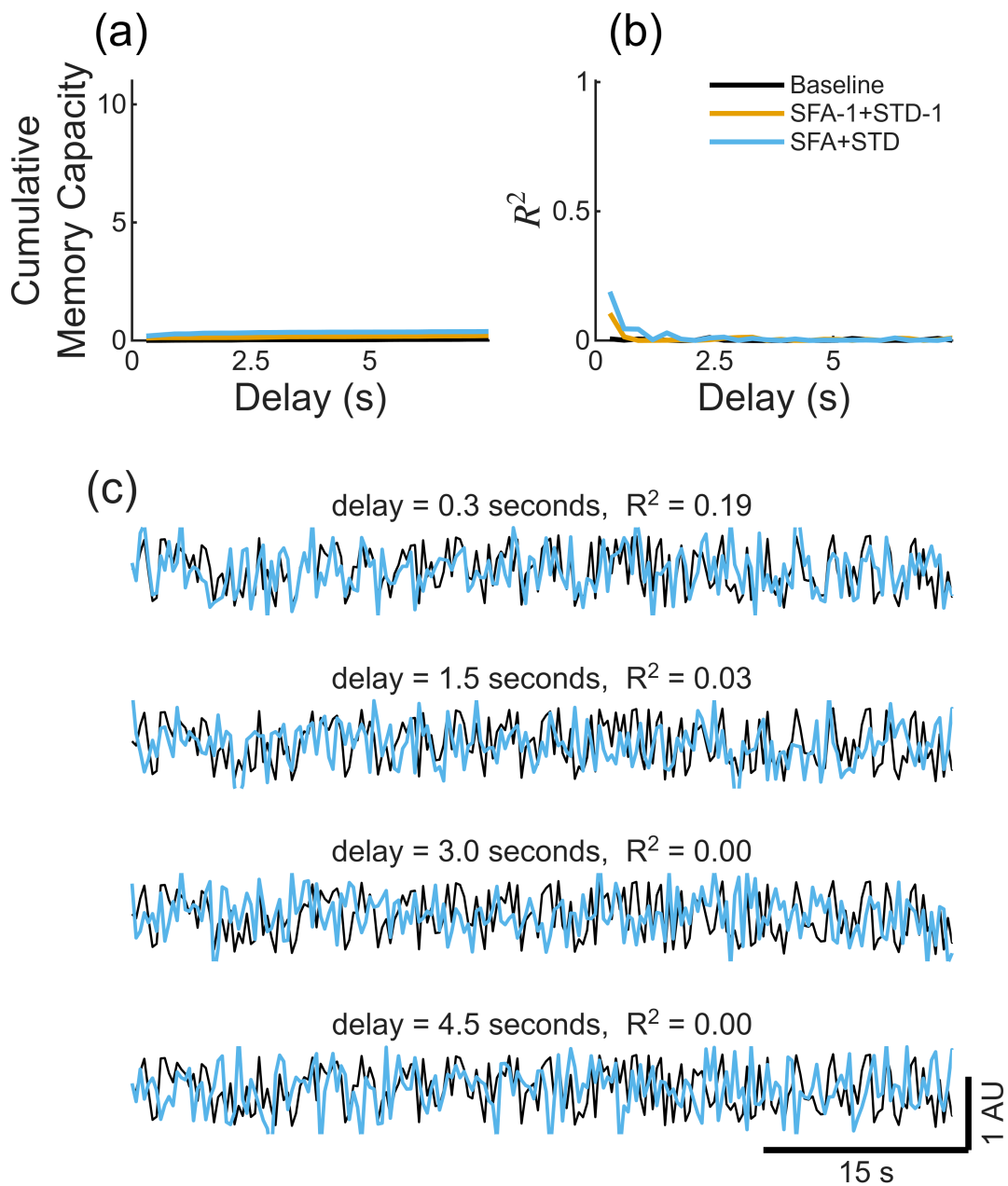


Figure 13: fig_memory_capacity_example/Fig_MC_Example.png

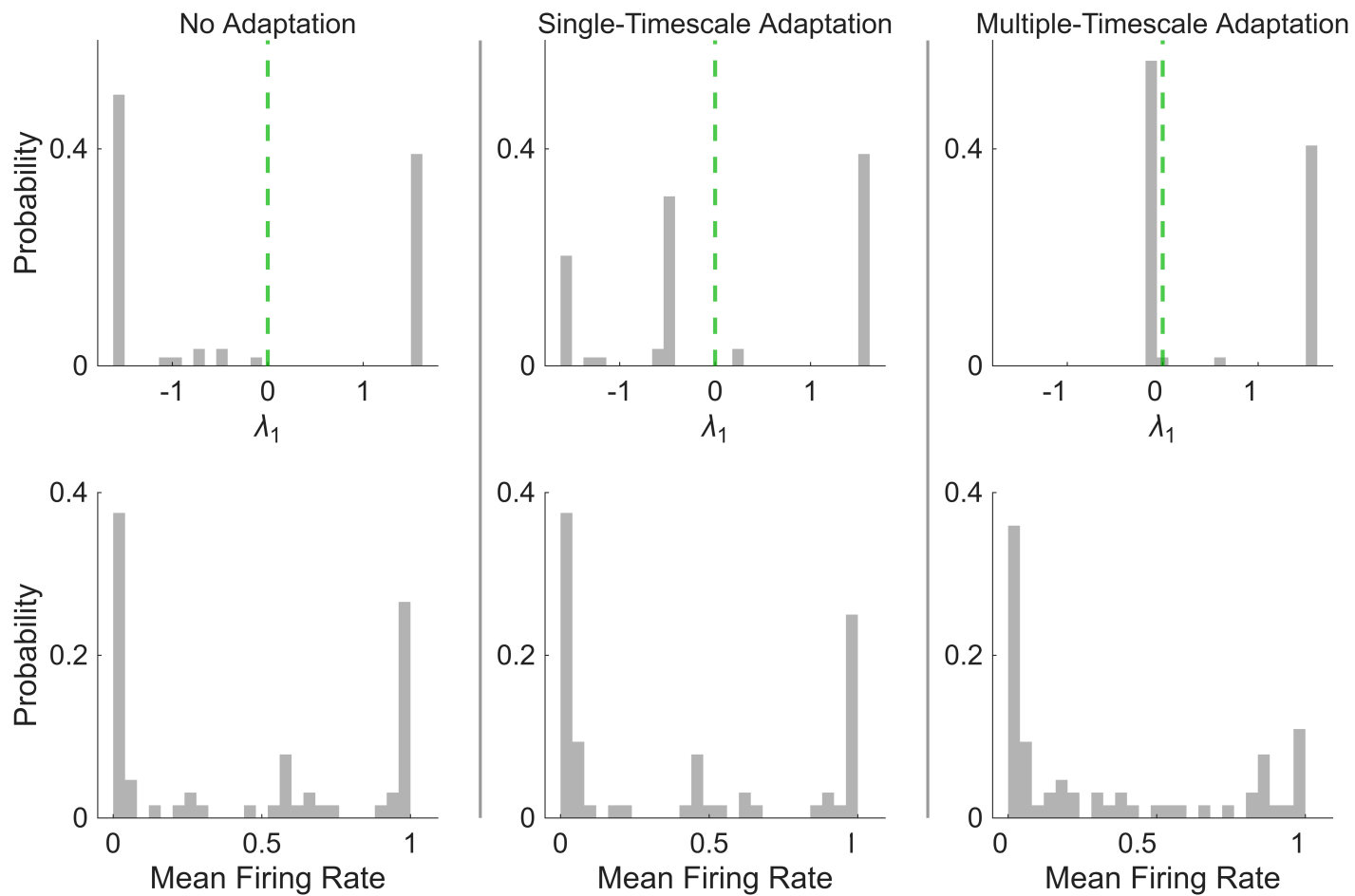


Figure 14: fig_param_space_allStd/Fig_ParamSpace_allStd.png

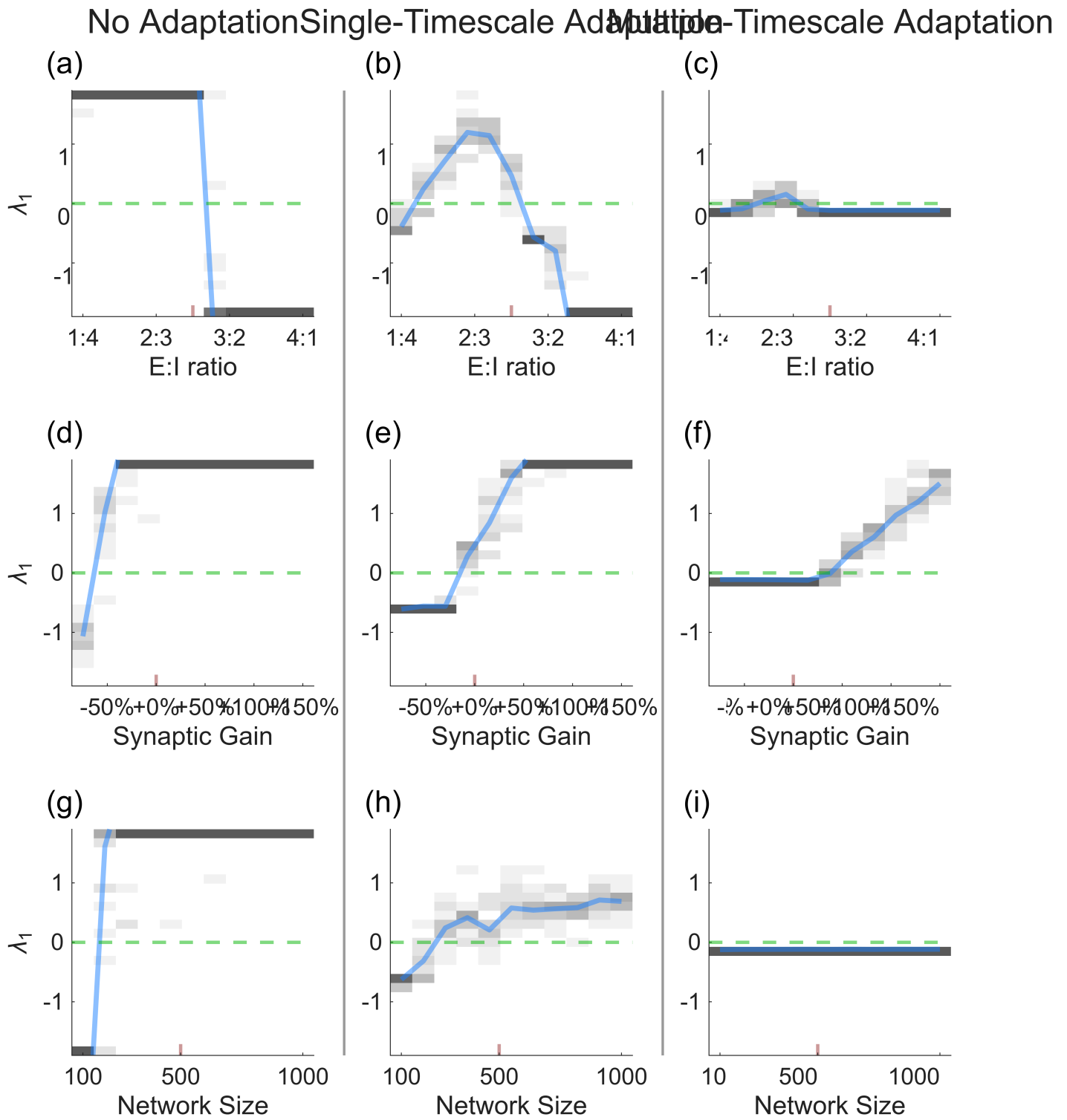


Figure 15: fig_sensitivity_analysis_allStd/Fig_Sensitivity_LLE_core.png

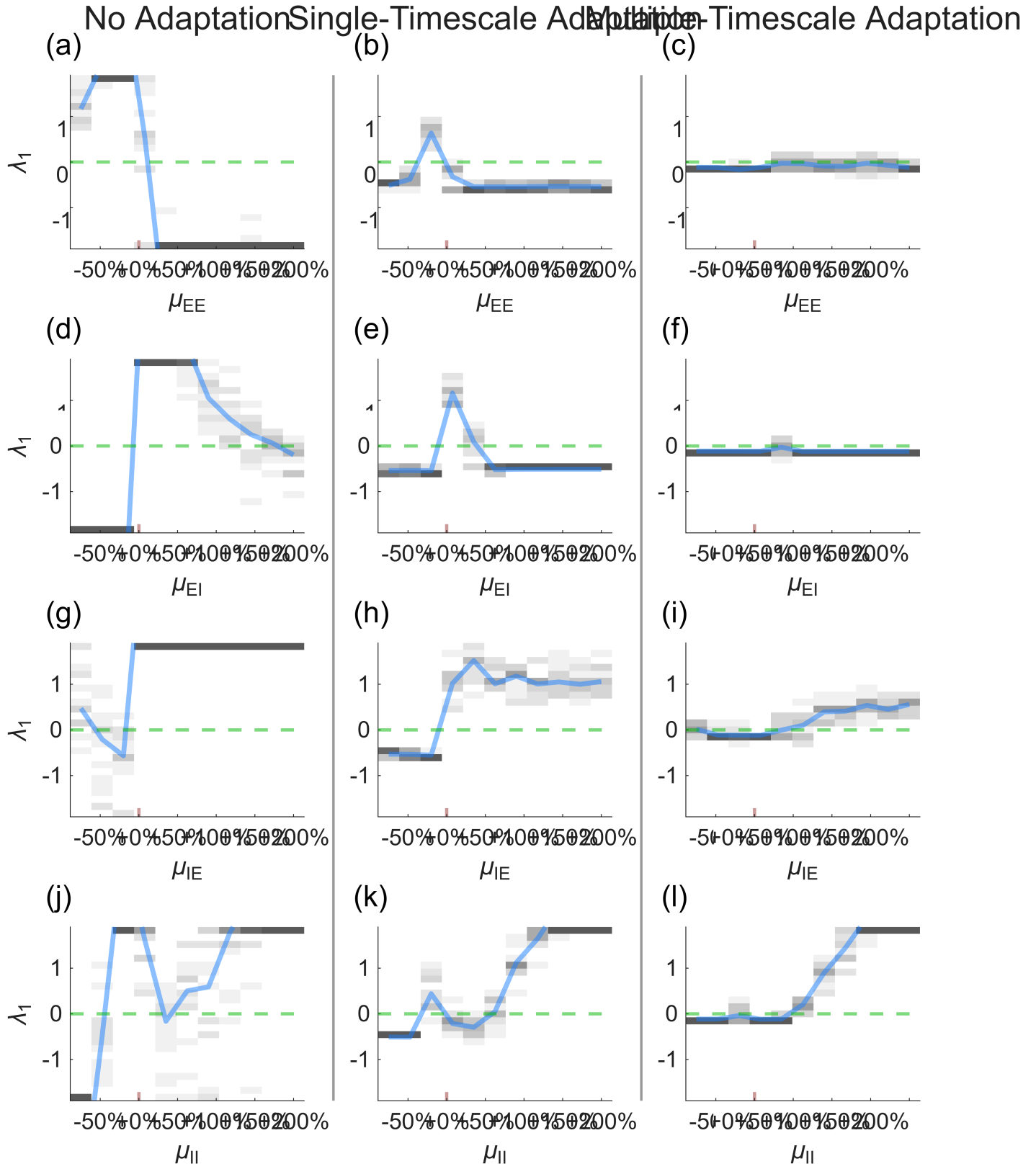


Figure 16: fig_sensitivity_analysis_allStd/fig_sensitivity_LLE_mu.png

No Adaptation Single-Timescale Adaptation Multiple-Timescale Adaptation

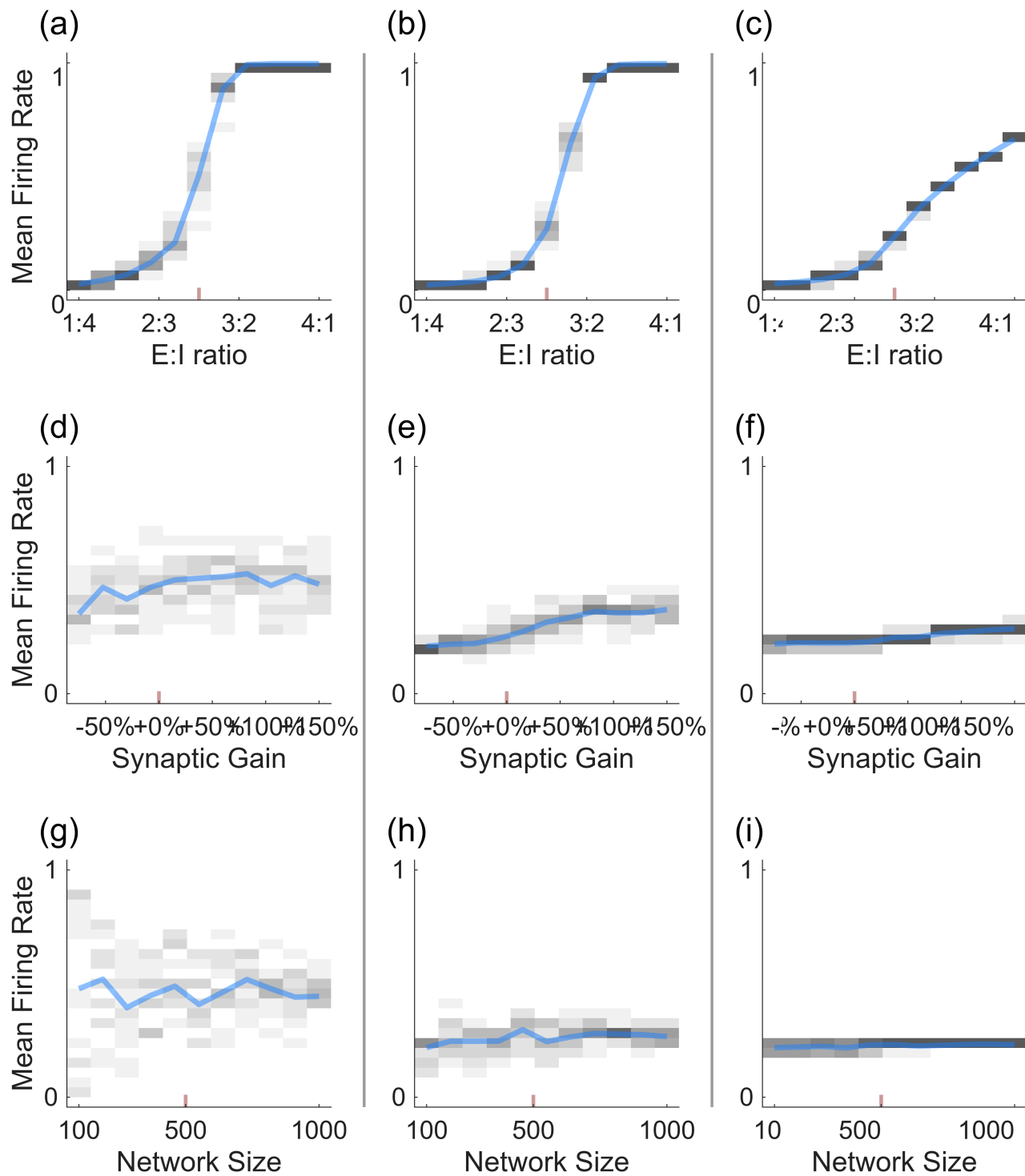


Figure 17: fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_core.png

fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_mu.png

fig_sensitivity_medians

fig_sensitivity_medians/Fig_Sensitivity_LLE_medians.png

fig_sensitivity_medians/Fig_sensitivity_mean_rate_medians.png

fig_sfa_EOC_allStd

fig_sfa_EOC_allStd/Fig_SFA_EOC_allStd.png

fig_SFA_steady_state

fig_SFA_steady_state/Fig_SFA_steady_state.png

fig_STD_steady_state

fig_STD_steady_state/Fig_STD_steady_state.png

fig_stim_engages_adaptation

fig_stim_engages_adaptation/bursting_psd.png

fig_stim_engages_adaptation/bursting_synchrony.png

fig_stim_engages_adaptation/bursting_timeseries.png

Regenerate with `write_figure_report('C:\Users\m218089\Desktop\github_repos\FractionalReservoir\figs\single_`
`figs/` is gitignored; this report references the images in place rather than copying them, so it is only ever as
current as its generation stamp above.

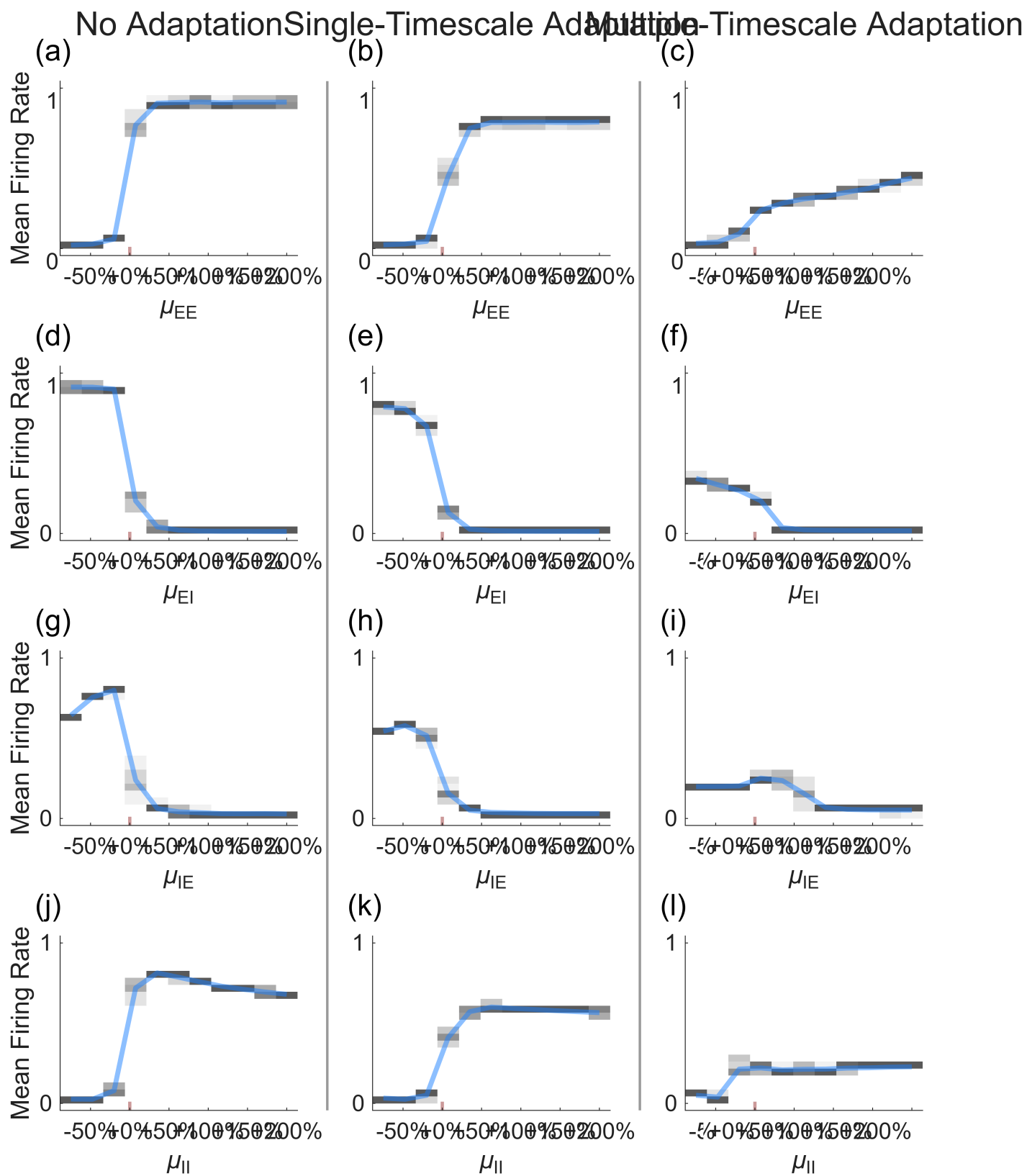


Figure 18: fig_sensitivity_analysis_allStd/Fig_sensitivity_mean_rate_mu.png

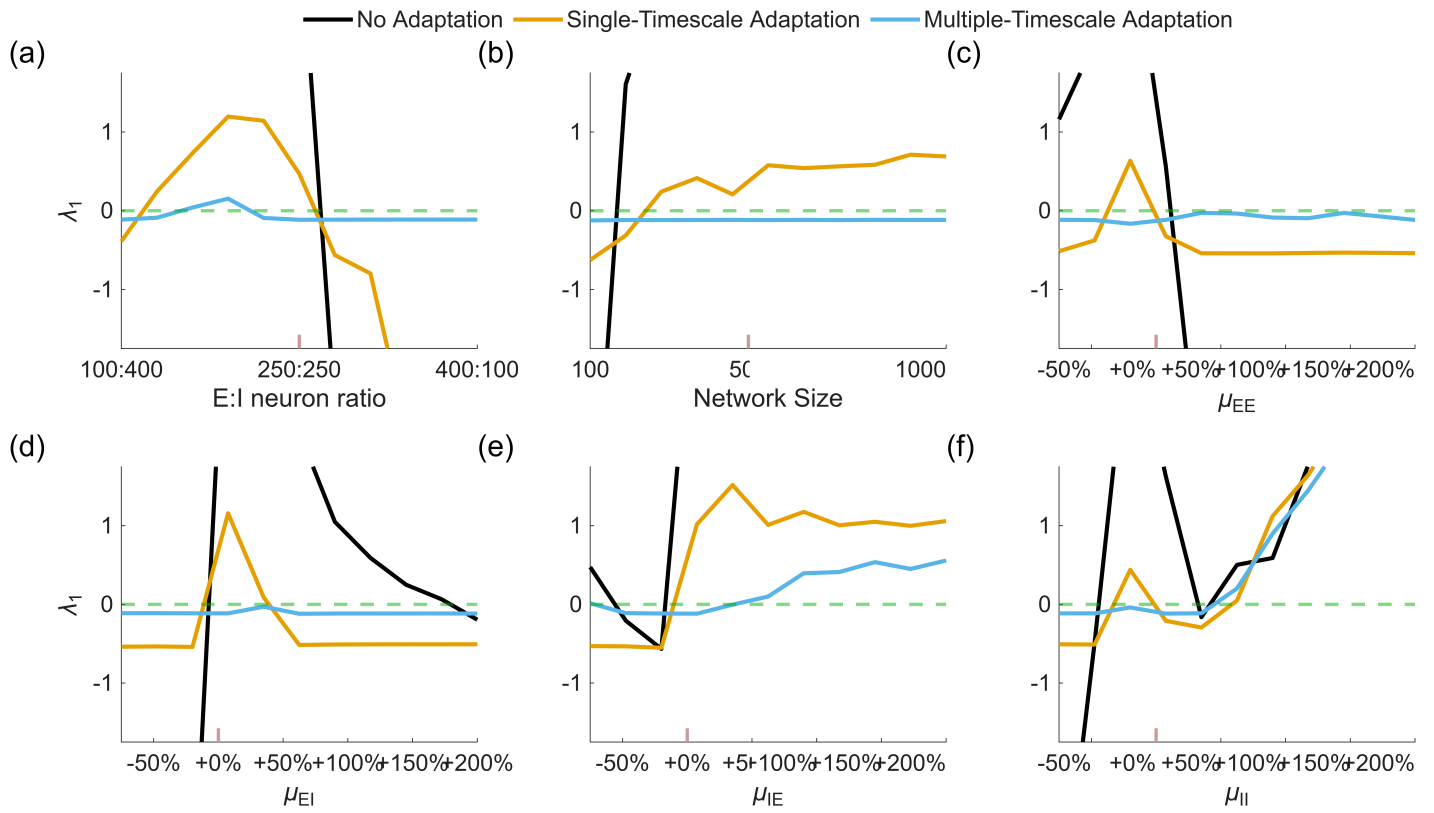


Figure 19: fig_sensitivity_medians/Fig_Sensitivity_LLE_medians.png

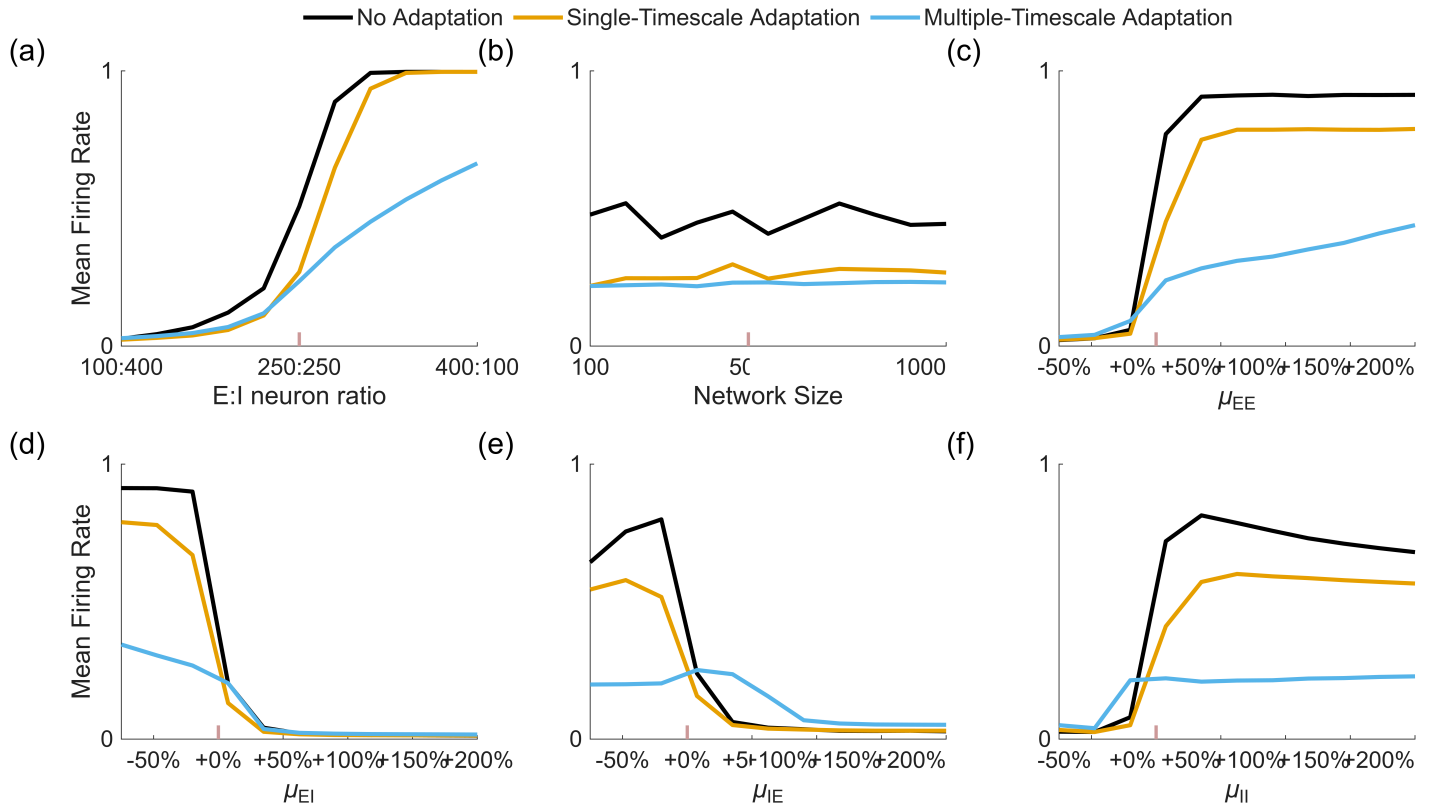


Figure 20: fig_sensitivity_medians/Fig_sensitivity_mean_rate_medians.png

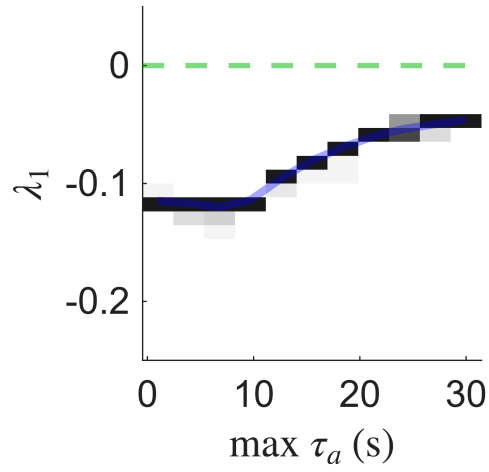


Figure 21: fig_sfa_EOC_allStd/Fig_SFA_EOC_allStd.png

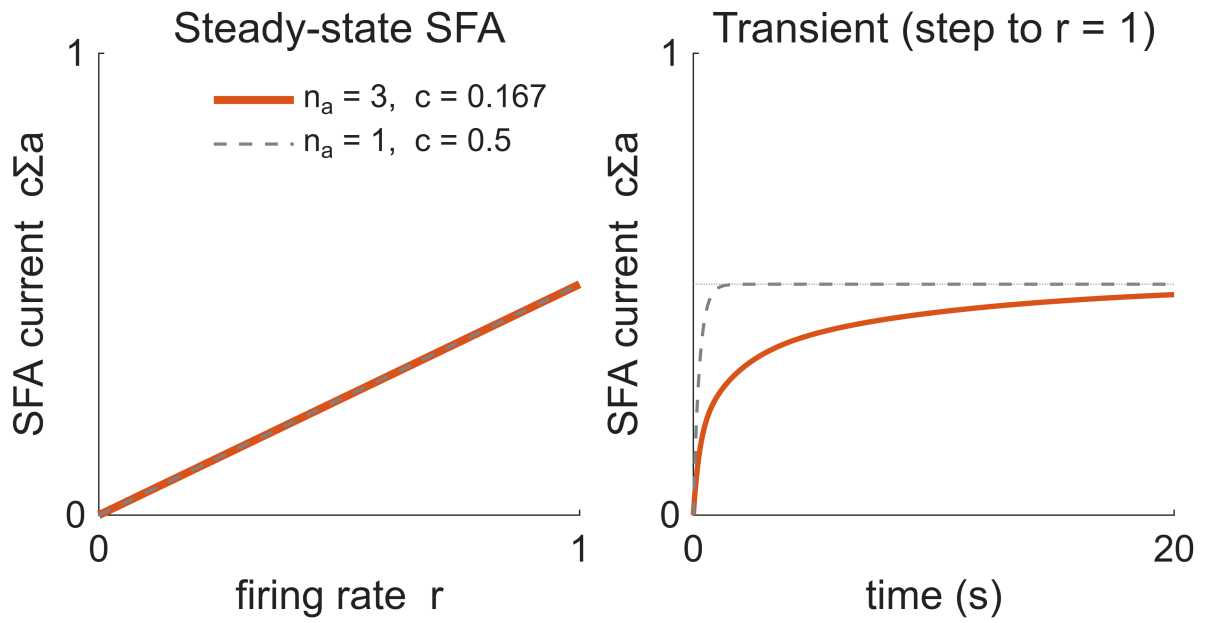


Figure 22: fig_SFA_steady_state/Fig_SFA_steady_state.png

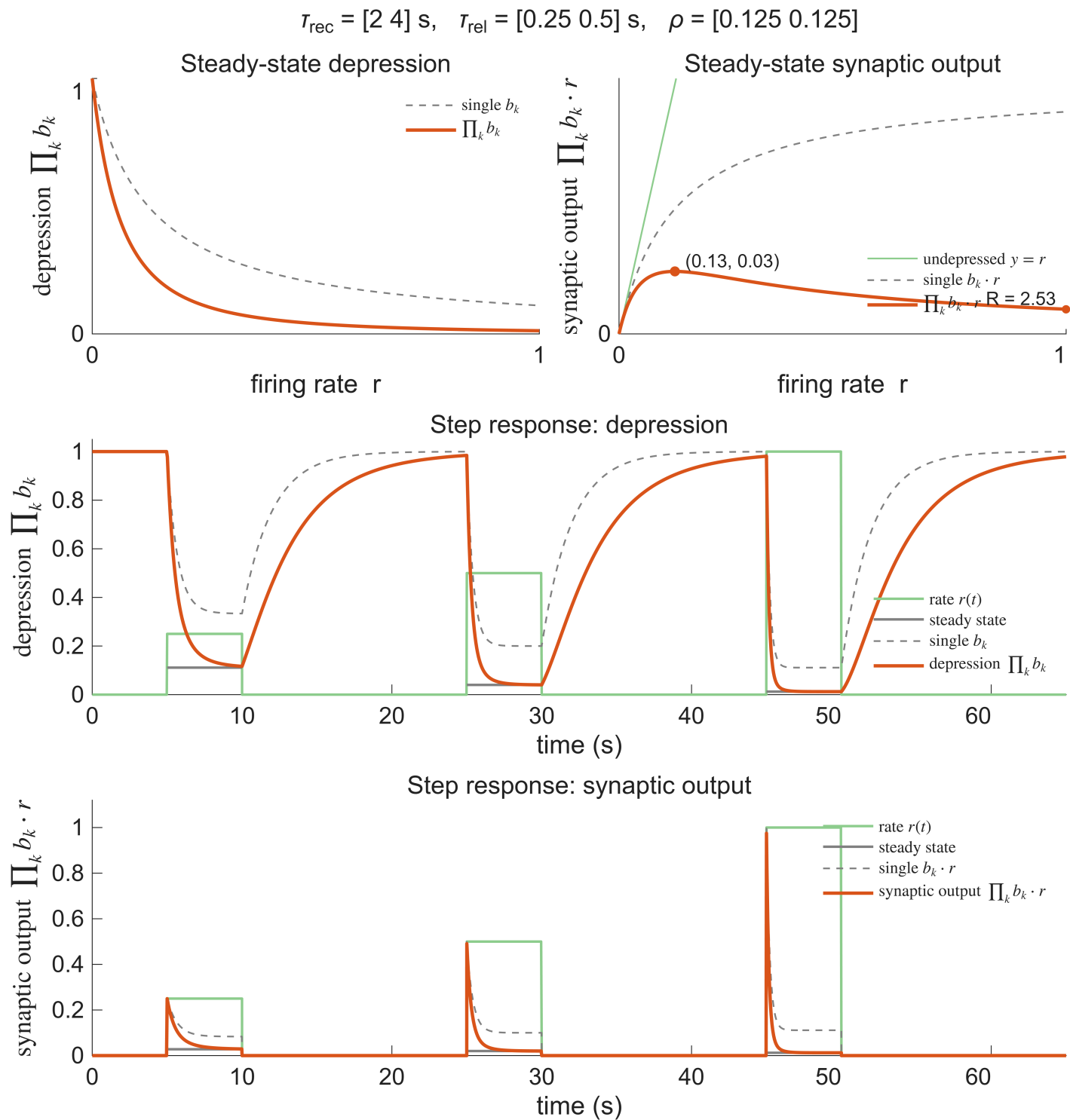


Figure 23: fig_STD_steady_state/Fig_STD_steady_state.png

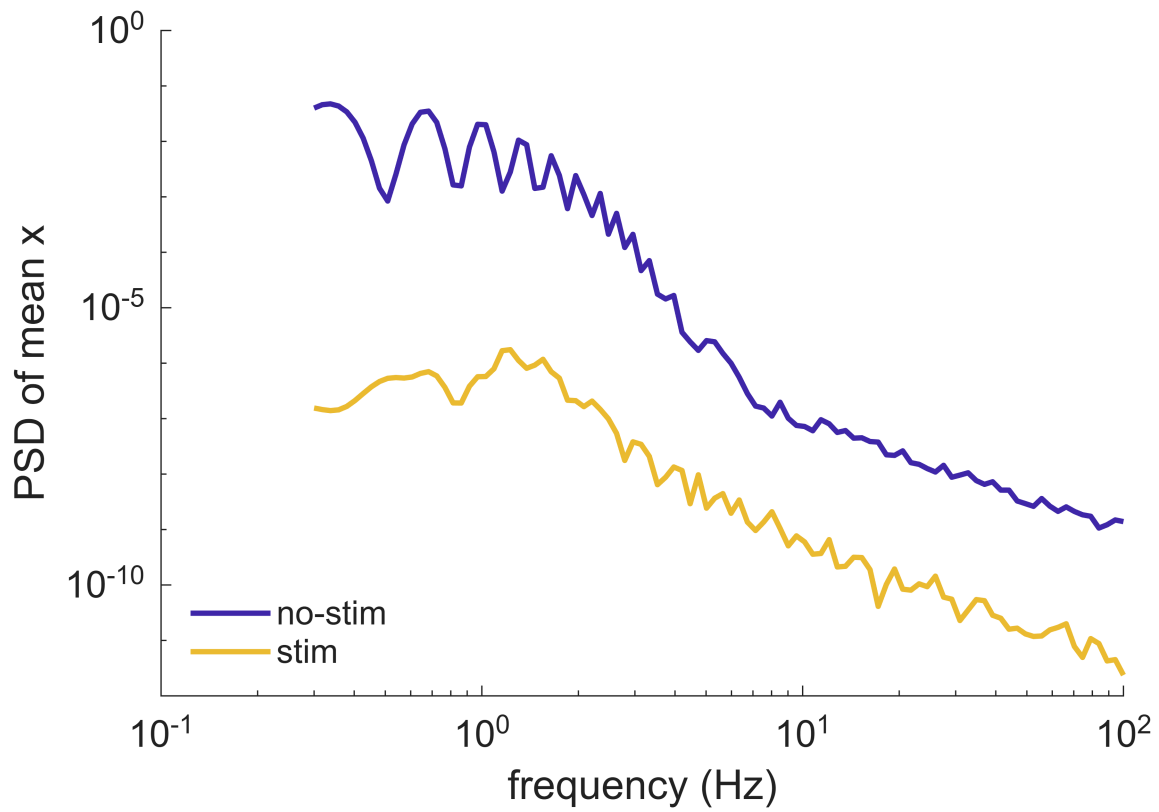


Figure 24: fig_stim_engages_adaptation/bursting_psd.png

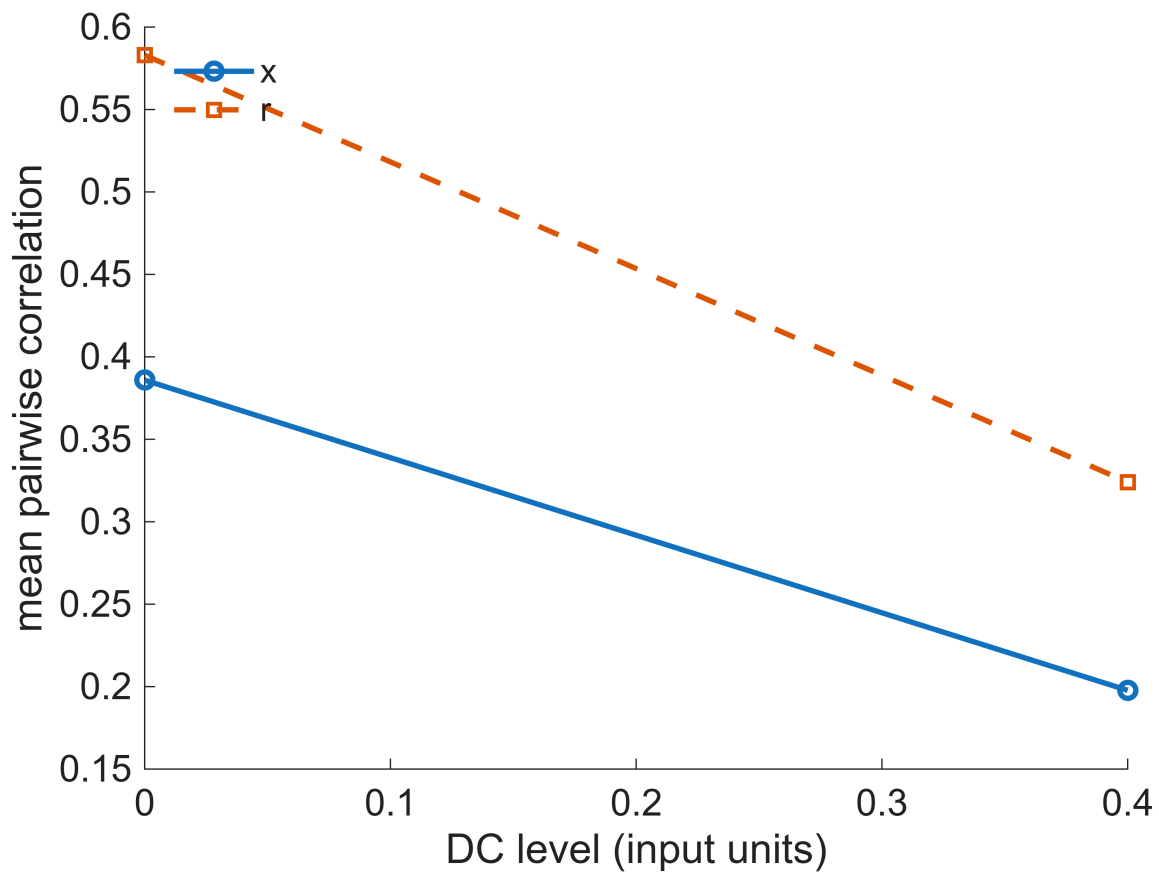


Figure 25: fig_stim_engages_adaptation/bursting_synchrony.png

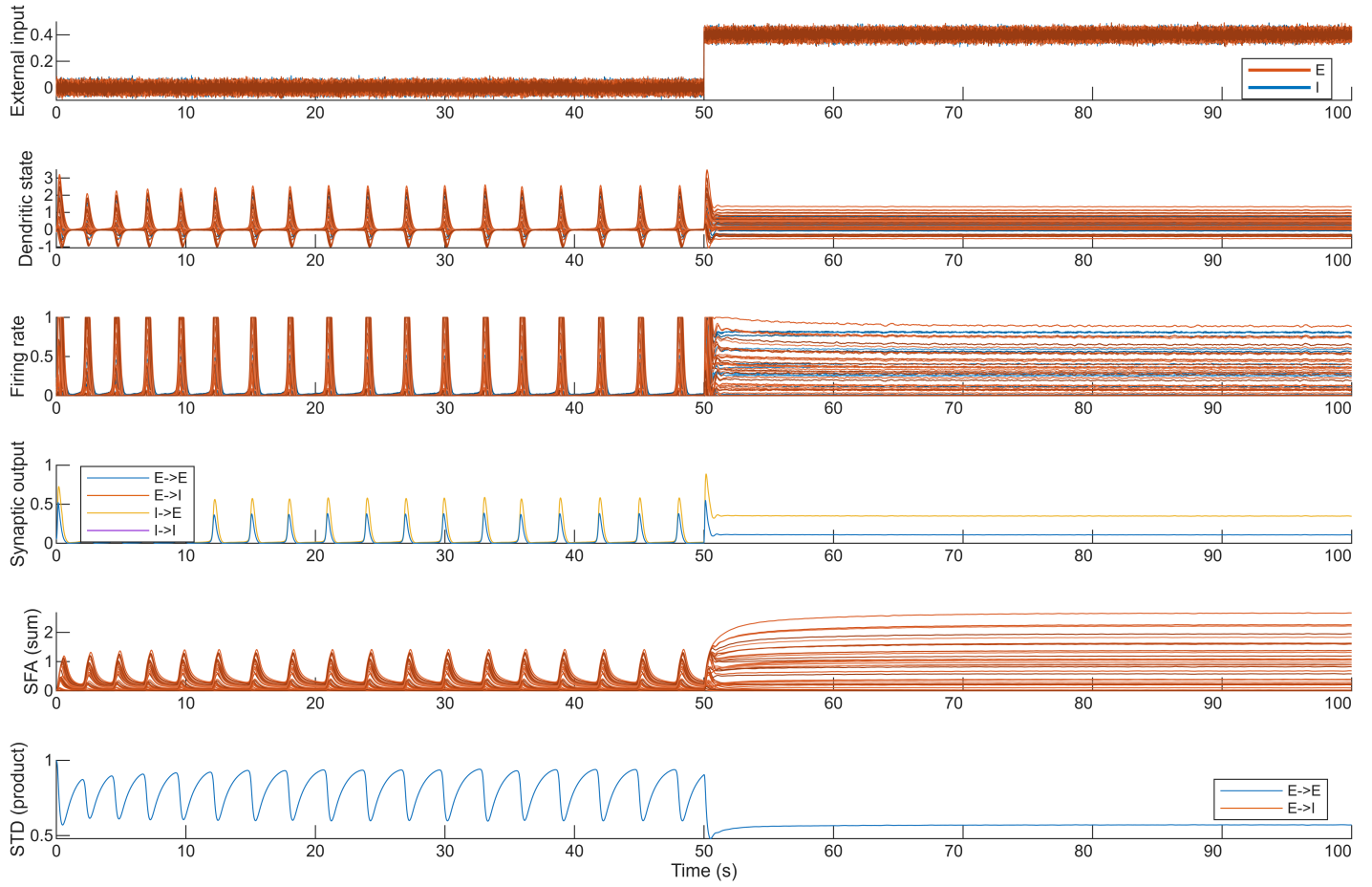


Figure 26: fig_stim_engages_adaptation/bursting_timeseries.png